LLM, Fine-tuning, RAG, and AI Agents

Study Notes

1. What is an LLM?

LLM stands for Large Language Model. It is an AI model trained to understand, generate, and work with human language (such as English or Hindi). It is called “large” because it has many parameters—the weights and biases learned during training. LLMs are trained on large collections of text, such as books, websites, and code.

Common tasks:

- Answering questions
- Writing paragraphs and essays
- Translating languages
- Writing code
- Conversational assistance
- Summarizing documents
- Customer support

Basic flow: Input → LLM → Output

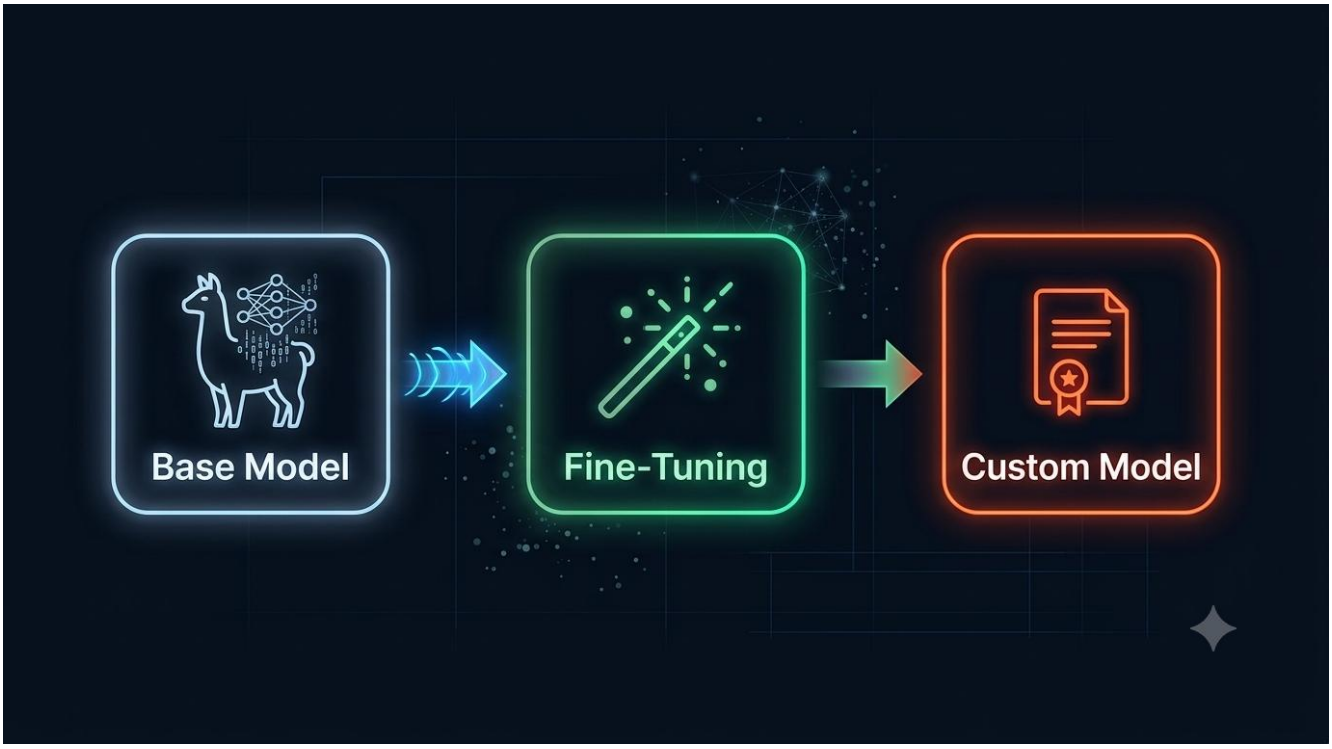


2. What is Fine-tuning?

Fine-tuning means taking a pre-trained model and training it further on a smaller, task- or domain-specific dataset so it performs better for a particular use case.

Analogy: Imagine hiring someone who already knows English, Hindi, and basic science. You train them further in medical topics so they become more specialized, without teaching everything from scratch.

Flow: Base Model → Fine-tuning → Custom Model



3. What is RAG?

RAG stands for Retrieval-Augmented Generation. It follows the idea: “Search first, then answer.” A retrieval system finds relevant information, and the LLM uses it to generate a response.

Typical flow: User question → Retriever → Relevant documents → LLM → Final answer

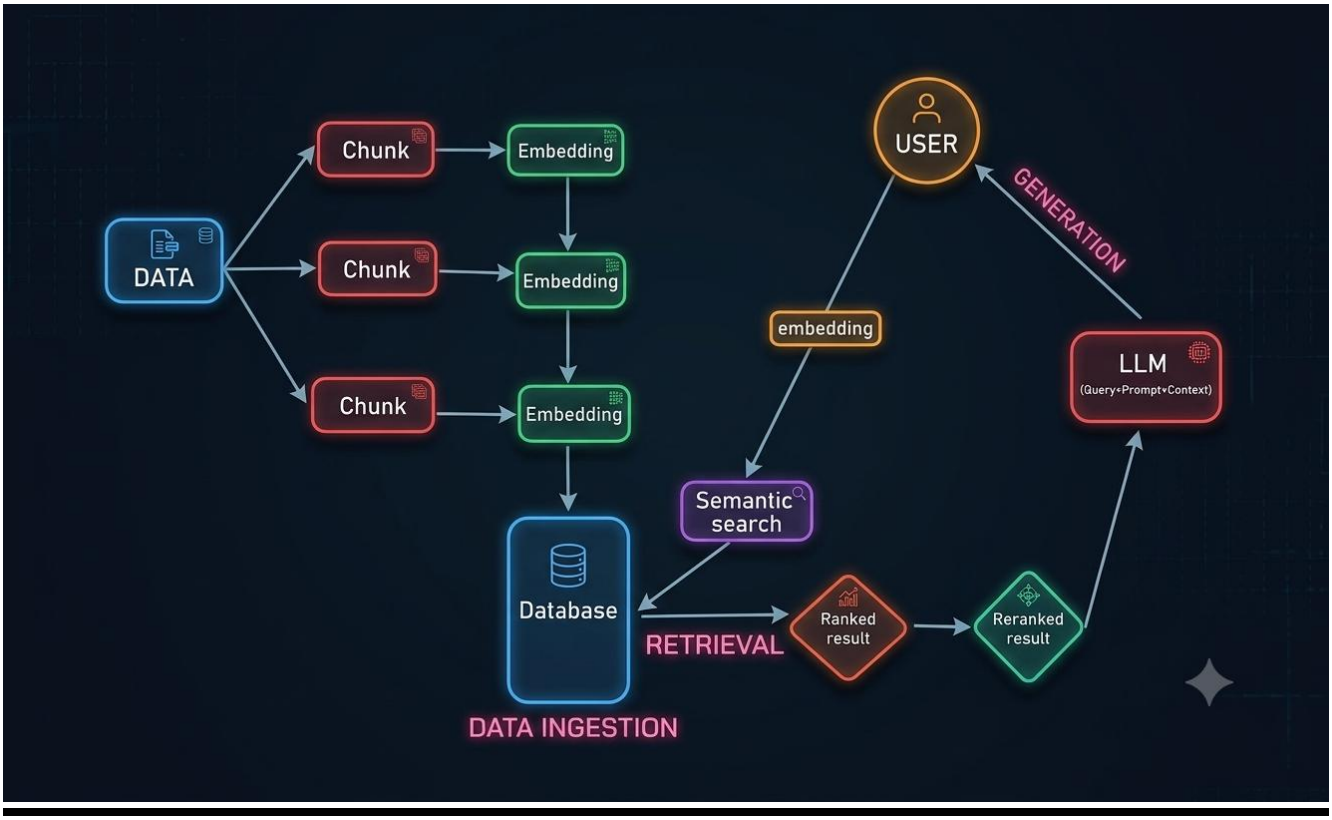
Example: If someone asks who won the 2022 FIFA World Cup, a RAG system can retrieve a trusted source stating that Argentina won, then use that information to answer

Retrieval + Generation = RAG

RAG architecture

Data ingestion: Source data → Chunks → Embeddings → Vector database

Answering: User question → Query embedding → Semantic search → Retrieved results → LLM → Response



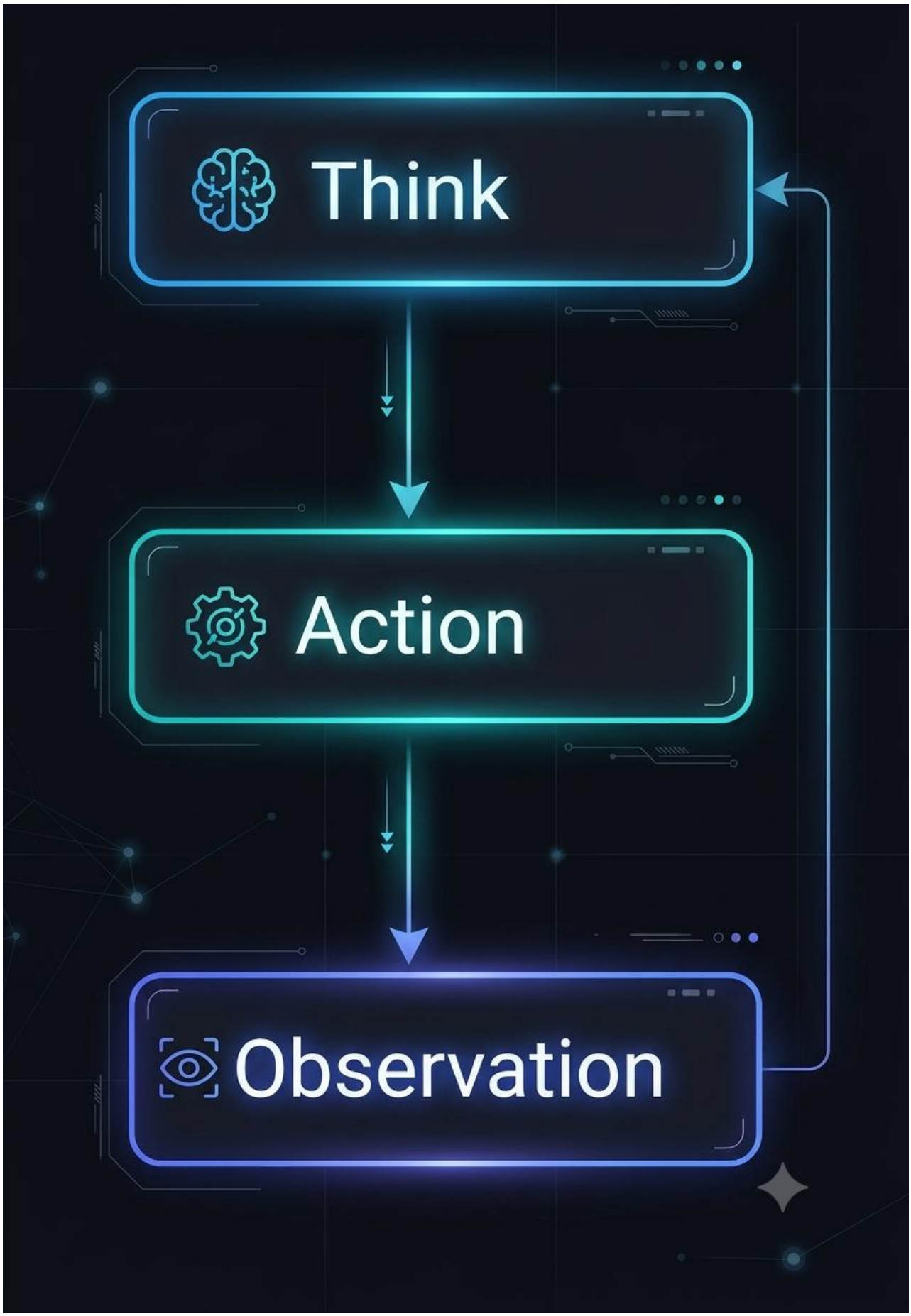
4. What are AI Agents?

An AI Agent is a system that uses an LLM to decide what to do and can use tools to perform actions toward a goal. A plain LLM can suggest places to visit, while an agent connected to suitable tools may search, compare, or book travel (with required permissions and confirmations).

Agent loop: Think / Plan → Action (tool call) → Observation → Next step

An agent may use:

- Instructions or system prompt
- Tools such as APIs, databases, or search
- Memory or state
- Planning and reasoning

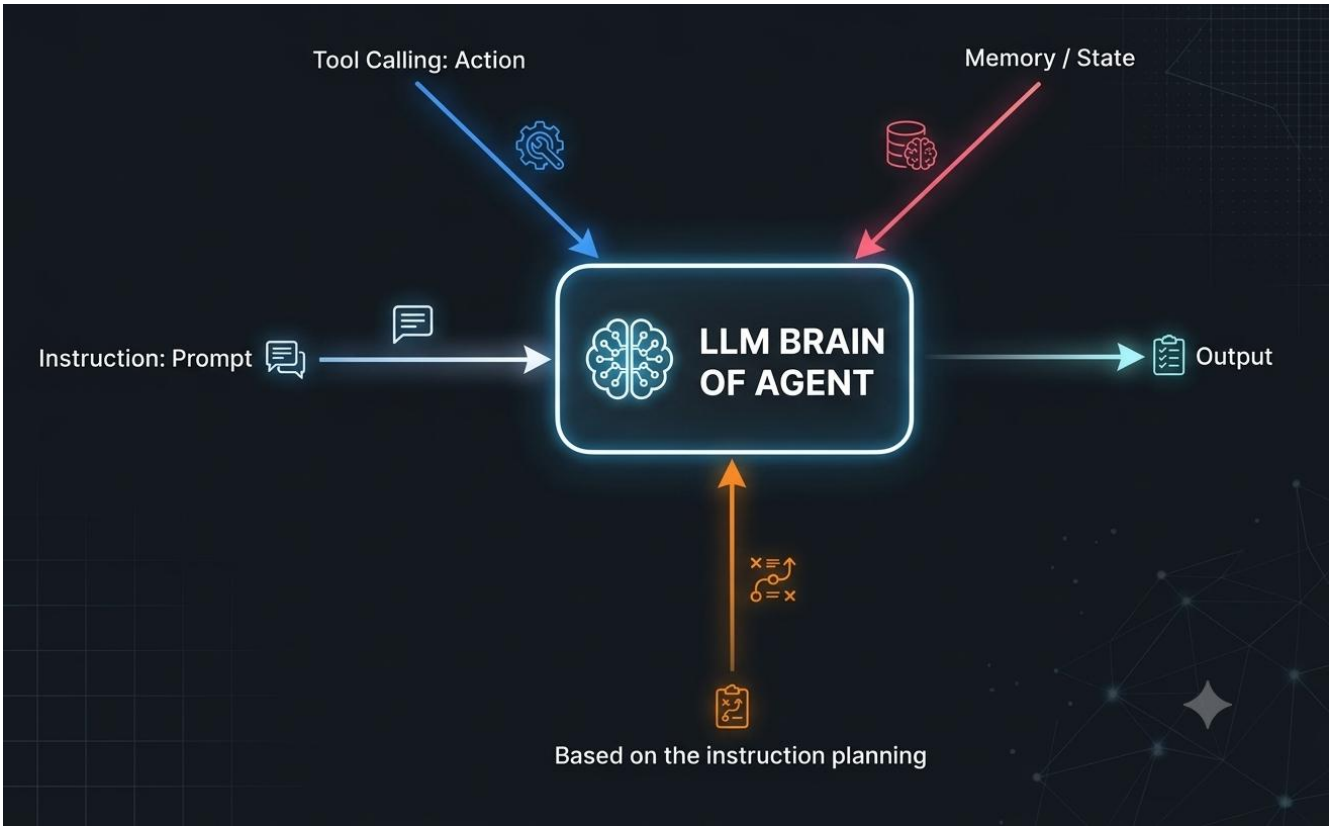

Think
Action
Observation

5. Fine-tuning vs RAG vs Agents

Approach	Purpose	Best suited for
LLM	General language generation	Chat, writing, summarization, simple Q&A
Fine-tuning	Adapt model behavior or task performance	Specialized style, format, or domain tasks
RAG	Supply relevant external knowledge at answer time	Fresh, private, or document-grounded answers
AI Agent	Plan and use tools to complete tasks	Multi-step workflows and permitted real-world actions

6. When should we use each?

Need	Recommended approach
Basic text generation, chat, writing, or simple Q&A	Use a pretrained LLM
Consistent domain behavior, style, or task-specific output	Fine-tune, if prompting alone is insufficient
Answers grounded in current or private documents	Use RAG
Multi-step tasks involving tools or external actions	Build an agent with tools and safeguards



7. Can they be combined?

Yes. A practical system can combine all three:

- Fine-tuned model: improves domain behavior, tone, or task performance.
- RAG layer: retrieves fresh or organization-specific information.
- Agent layer: chooses steps and uses tools to complete tasks.

Analogy:

Fine-tuning = expert brain;

RAG = access to books or search;

Agent = a person who can use tools to act.